



PROJECT REPORT ON  
**“Max Electronics Admin Hub: Managing Your Online Store”**

IN PARTIAL FULFILLMENT OF

BACHELOR OF COMPUTER SCIENCE

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PROJECT GUIDE

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Subject: Advance Database Management System

## **1. Introduction:**

The project serves as the administrative dashboard for an electronic shop website. It provides the shop's administrators with tools and functionalities to manage the website's content, including products, categories, pricing, and stocks. This admin panel ensures efficient control over the online store's operations.

## **2. Objectives:**

**Product Management:** Enable administrators to add, edit, and remove products, including details such as name, category, price, and stock quantity.

**Category Management:** Allow administrators to create and manage product categories for better organization.

**Database Integration:** Utilize MySQL (Pl/Sql) for storing and managing product

**User-Friendly Interface:** Create an intuitive and responsive web interface for administrators.

**Scalability:** Design the system to handle growth and adapt to future enhancements.

## **3. System Requirements:**

**Server:** GlassFish Server 4.1.1

**Java:** JDK (Java Development Kit) 8.

**Development Tools:** Netbeans IDE 8.2.

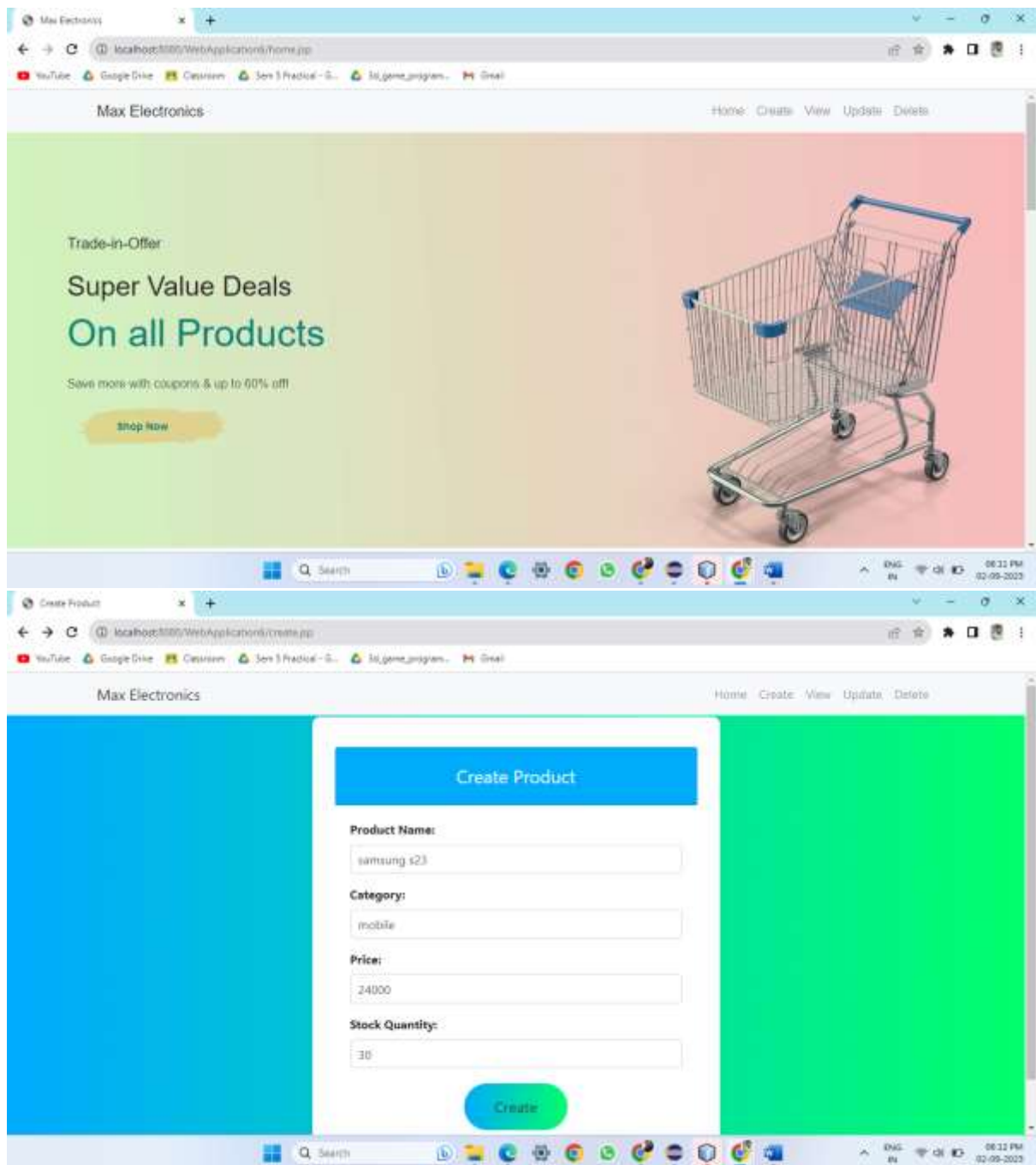
**Database:** MySQL (PL/SQL).

**Web Technologies:** JSP, Servlets, HTML, CSS.

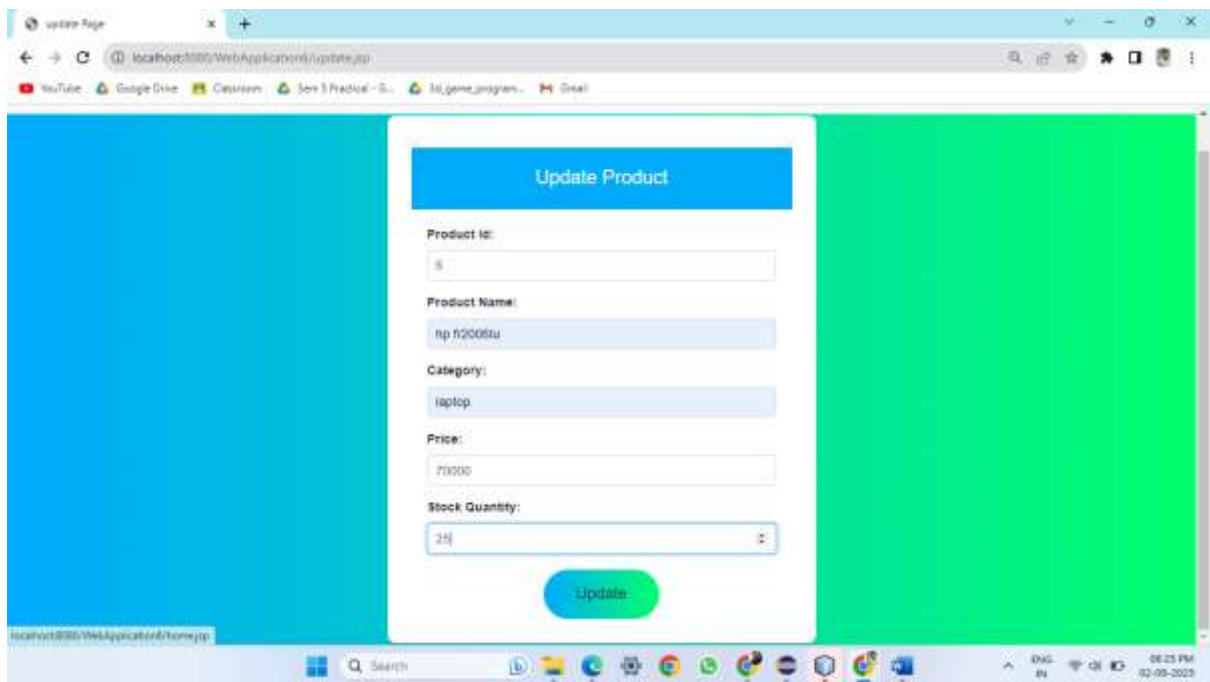
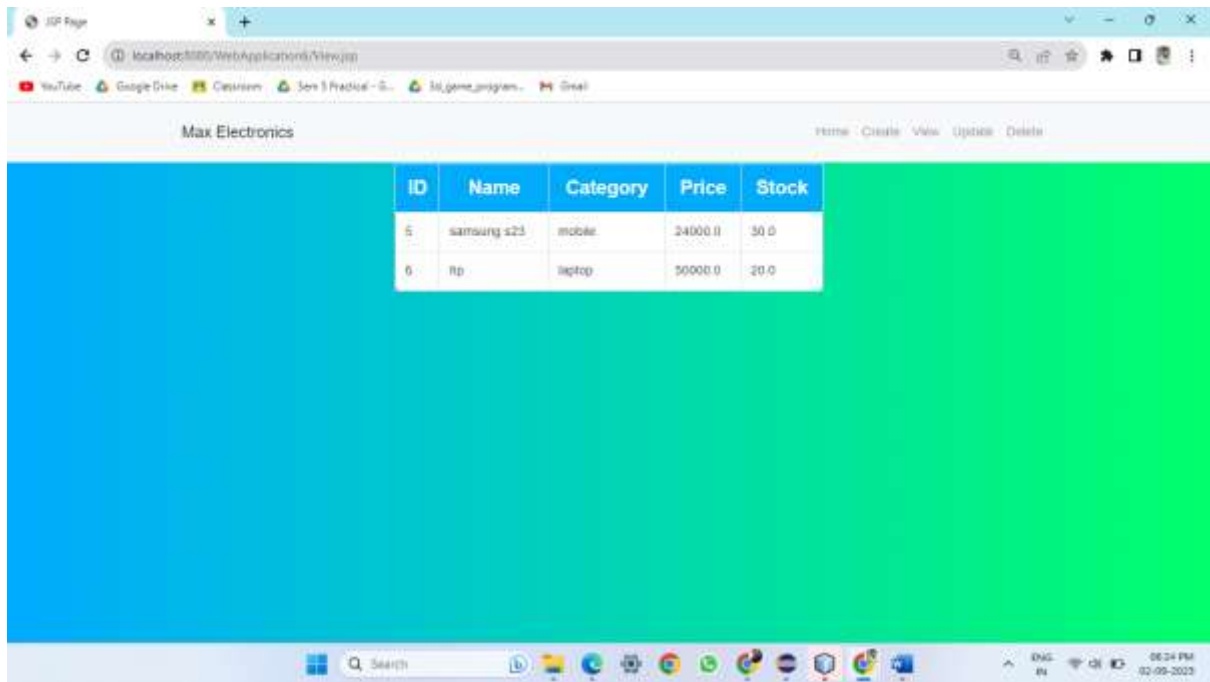
**Database Connectivity:** JDBC (Java Database Connectivity) driver for MySQL.

**Web Browser:** Chrome, Firefox, or any modern browser.

#### 4. Code with Output Overview:



Product created successfully



Product details update successfully

| ID | Name        | Category | Price   | Stock |
|----|-------------|----------|---------|-------|
| 5  | samsung s23 | mobile   | 24000.0 | 30.0  |
| 6  | hp fr2006tu | laptop   | 70000.0 | 25.0  |

Max Electronics

Home | Create | View | Update | Delete

### Delete Product

Enter Product id:

Delete

Product deleted successfully

| ID | Name        | Category | Price   | Stock |
|----|-------------|----------|---------|-------|
| 6  | hp fr2006tu | laptop   | 70000.0 | 25.0  |

## Code:

Userdao.java

```
public class UserDao {  
  
    private static final String URL = "jdbc:mysql://localhost:3308/adbms";  
    private static final String USERNAME = "root";  
    private static final String PASSWORD = "root75";  
  
    public static Connection getConnection() {  
        Connection connection = null;  
        try {  
            Class.forName("com.mysql.jdbc.Driver");  
            connection = DriverManager.getConnection(URL, USERNAME, PASSWORD);  
        } catch (ClassNotFoundException | SQLException e) {  
        }  
        return connection;  
    }  
}
```

Registeruser.java

```
@WebServlet("/registeruser")  
public class registeruser extends HttpServlet {  
    @Override  
    protected void doPost(HttpServletRequest request, HttpServletResponse response)  
        throws ServletException, IOException {  
        response.setContentType("text/html;charset=UTF-8");  
        try (PrintWriter out = response.getWriter()) {  
            String name = request.getParameter("fullname");  
            String email = request.getParameter("email");  
            String password = request.getParameter("passsword");  
            RequestDispatcher dispatcher = null;
```

```

Connection conn = null;

try{
conn = UserDao.getConnection();
String sql = "{CALL registerUser(?, ?, ?)}";
CallableStatement cstmt = conn.prepareCall(sql);
cstmt.setString(1, name);
cstmt.setString(2, email);
cstmt.setString(3, password);

int Result=cstmt.executeUpdate();

    dispatcher = request.getRequestDispatcher("login.jsp");
    if(Result > 0){
        request.setAttribute("status","success");
    }
    else{
        request.setAttribute("status","failed");
    }
    dispatcher.forward(request,response);
}
catch(Exception e){
    out.println(e);
}
}
}

```

LoginServlet.java

```

@WebServlet(name = "login", urlPatterns = {"/login"})

public class LoginServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    @Override

```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
    throws ServletException, IOException {
```

```
    response.setContentType("text/html;charset=UTF-8");
```

```
    String email = request.getParameter("login-email");
```

```
    String password = request.getParameter("login-password");
```

```
    Connection conn = null;
```

```
    PreparedStatement pstmt = null;
```

```
    ResultSet rs = null;
```

```
    try {
```

```
        conn = UserDao.getConnection();
```

```
        String sql = "SELECT GetUserPasswordByEmail(?) AS password";
```

```
        pstmt = conn.prepareStatement(sql);
```

```
        pstmt.setString(1, email);
```

```
        rs = pstmt.executeQuery();
```

```
        if (rs.next()) {
```

```
            String dbPassword = rs.getString("password");
```

```
            if (dbPassword.equals(password)) {
```

```
                response.sendRedirect("home.jsp");
```

```
                return;
```

```
            }
```

```
        }
```

```
        PrintWriter out = response.getWriter();
```

```
        out.println("Login Failed");
```

```
    } catch (SQLException e) {
```

```
        e.printStackTrace();
```

```
    } finally {
```

```
        try {
```

```
            if (rs != null) rs.close();
```

```
            if (pstmt != null) pstmt.close();
```

```

        if (conn != null) conn.close();
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}
}
}

```

Createservlet.java

```
import dao.UserDao;
```

```
@WebServlet(name = "createservlet", urlPatterns = {"/createservlet"})
```

```
public class createservlet extends HttpServlet {
```

```
    private static final long serialVersionUID = 1L;
```

```
    @Override
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        response.setContentType("text/html;charset=UTF-8");
```

```
        String productname = request.getParameter("productname");
```

```
        String category = request.getParameter("category");
```

```
        int price = Integer.parseInt(request.getParameter("price"));
```

```
        int stockquantity = Integer.parseInt(request.getParameter("stockquantity"));
```

```
        try {
```

```
            Connection conn = UserDao.getConnection();
```

```
            CallableStatement cstmt = null;
```

```
            String sql = "{CALL createproduct(?, ?, ?, ?)}";
```

```
            cstmt = conn.prepareCall(sql);
```

```
            cstmt.setString(1, productname);
```

```

        pstmt.setString(2, category);
        pstmt.setInt(3, price);
        pstmt.setInt(4, stockquantity);
        int result = pstmt.executeUpdate();

        if (result > 0) {
            request.setAttribute("message", "Product created successfully");
        } else {
            request.setAttribute("message", "Invalid ID or character");
        }

        pstmt.close();
        conn.close();
    } catch (SQLException e) {
        e.printStackTrace();
        request.setAttribute("message", "Error: " + e.getMessage());
    }
    request.getRequestDispatcher("/create.jsp").forward(request, response);
}
}

```

## View.jsp

```

<%
    try {
        Connection conn = UserDao.getConnection();
        String sql = "{CALL getAllProducts()}";
        CallableStatement pstmt = conn.prepareCall(sql);
        ResultSet rs = pstmt.executeQuery();

        while (rs.next()) {
            int id = rs.getInt("id");

```

```

        String name = rs.getString("name");
        String category = rs.getString("category");
        double price = rs.getDouble("price");
        double stock_quantity = rs.getDouble("stock_quantity");
        out.println("<tr>");
        out.println("<td>" + id + "</td>");
        out.println("<td>" + name + "</td>");
        out.println("<td>" + category + "</td>");
        out.println("<td>" + price + "</td>");
        out.println("<td>" + stock_quantity + "</td>");
        out.println("</tr>");
    }
    rs.close();
    pstmt.close();
    conn.close();
} catch (Exception e) {
    e.printStackTrace();
}
%>

```

## Updateservlet.java

```

import dao.UserDao;

@WebServlet("/UpdateServlet")

public class UpdateServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();

        try {

```

```

int id = Integer.parseInt(request.getParameter("id"));
String productname = request.getParameter("productname");
String category = request.getParameter("category");
int price = Integer.parseInt(request.getParameter("price"));
int stockquantity = Integer.parseInt(request.getParameter("stockquantity"));

Connection conn = UserDao.getConnection();
String sql = "{CALL UpdateProduct(?, ?, ?, ?,?)}";
CallableStatement cstmt = conn.prepareCall(sql);
cstmt.setInt(1, id);
cstmt.setString(2, productname);
cstmt.setString(3, category);
cstmt.setInt(4, price);
cstmt.setInt(5, stockquantity);
int rowsUpdated = cstmt.executeUpdate();
cstmt.close();
conn.close();

if (rowsUpdated > 0) {
    request.setAttribute("message", "Product details update successfully");
} else {
    request.setAttribute("message", "Update failed");
}

request.getRequestDispatcher("update.jsp").forward(request, response);

} catch (Exception e) {
    e.printStackTrace();
    request.setAttribute("message", "An error occurred");
    request.getRequestDispatcher("update.jsp").forward(request, response);
}

```

```
    }  
}  
}
```

## **Deleteservlet.java**

```
@WebServlet("/DeleteServlet")  
  
public class DeleteServlet extends HttpServlet {  
    private static final long serialVersionUID = 1L;  
  
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws  
ServletException, IOException {  
        response.setContentType("text/html;charset=UTF-8");  
  
        try {  
            int id = Integer.parseInt(request.getParameter("id"));  
            Connection conn = UserDao.getConnection();  
            String sql = "{CALL deleteById(?)}";  
            CallableStatement cstmt = conn.prepareCall(sql);  
            cstmt.setInt(1, id);  
            cstmt.executeUpdate();  
            cstmt.close();  
            conn.close();  
  
            String message = "Product deleted successfully";  
            request.setAttribute("message", message);  
            request.getRequestDispatcher("delete.jsp").forward(request, response);  
        } catch (Exception e) {  
            e.printStackTrace();  
        }  
    }  
}
```

## 5)Procedures and fucnctions using plsql in mysql

### **GetUserPasswordByEmail**

```
DELIMITER @@  
  
CREATE FUNCTION adbms.GetUserPasswordByEmail  
(p_email VARCHAR(50))  
RETURNS varchar(20) CHARSET latin1  
BEGIN  
  
    DECLARE v_password VARCHAR(20);  
  
    SELECT password INTO v_password  
    FROM users  
    WHERE email = p_email;  
  
    RETURN v_password;  
  
END @@  
  
DELIMITER ;
```

### **UpdateProduct**

```
DELIMITER @@  
  
CREATE PROCEDURE adbms.UpdateProduct  
(  
    IN p_id INT,  
    IN p_name VARCHAR(100),  
    IN p_category VARCHAR(50),  
    IN p_price DECIMAL(10, 2),  
    IN p_stock_quantity DECIMAL(10, 2)  
)  
BEGIN  
  
    UPDATE products  
    SET name = p_name,
```

```
        category = p_category,  
        price = p_price,  
        stock_quantity = p_stock_quantity  
WHERE id = p_id;  
END @@  
DELIMITER ;
```

### **Createproduct**

```
DELIMITER @@  
CREATE PROCEDURE mysql.createproduct  
(  
    IN p_name VARCHAR(100),  
    IN p_category VARCHAR(50),  
    IN p_price DECIMAL(10, 2),  
    IN p_stock_quantity DECIMAL(10, 2)  
)  
BEGIN  
    INSERT INTO products (name, category, price, p_stock_quantity)  
    VALUES (p_name, p_category, p_price, p_stock_quantity);  
END @@  
DELIMITER ;
```

### **deleteById**

```
DELIMITER @@  
CREATE PROCEDURE adbms.deleteById  
(IN p_id INT)  
BEGIN  
    DELETE FROM products WHERE id = p_id;  
END @@
```

DELIMITER ;

### **registerUser**

DELIMITER @@

CREATE PROCEDURE adbms.registerUser

(

IN p\_username VARCHAR(50),

IN p\_email VARCHAR(50),

IN p\_password VARCHAR(20)

)

BEGIN

INSERT INTO users (username, email, password)

VALUES (p\_username, p\_email, p\_password);

END @@

DELIMITER ;

### **getAllProducts**

DELIMITER @@

CREATE PROCEDURE adbms.getAllProducts

()

BEGIN

SELECT \* FROM products;

END @@

DELIMITER ;